THOMAS NGUYEN

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PROFESSIONAL SUMMARY

Data science researcher and analytics professional with a Master of Data Science and Innovation from the University of Technology Sydney and recipient of the 2026 TD School Merit Award for Outstanding Academic Achievement. Author of two accepted research papers for publication at the 2026 Australasian Data Science and Machine Learning Conference (AusDM'26) and in the MDPI journal Risks, covering machine learning, cyber-risk modelling, probabilistic forecasting and actuarial risk assessment. Skilled in Python, SQL, Excel, Power BI, Tableau, statistical modelling and model validation, with practical experience in sports consultancy, data analysis and stakeholder-focused reporting.

TECHNICAL SKILLS

- **Data Analysis & Programming:** Python (Pandas, NumPy), SQL (joins, aggregations, subqueries, data transformation), Excel (PivotTables, XLOOKUP, SUMIF, Power Query)
- **Machine Learning & AI:** Model training and evaluation, feature engineering, cross-validation, hyperparameter tuning, neural networks, probabilistic modelling
- **Data Visualisation & Business Intelligence:** Power BI, Tableau, Excel dashboards and reporting
- **Statistical Analysis:** Descriptive statistics, hypothesis testing, correlation, regression, time-series analysis, multivariate analysis
- **Project & Collaboration Tools**: MS Office Suite, Trello, Jira

PROFESSIONAL EXPERIENCE

Sport Consultancy & Data Analyst Intern **September 2025 – December 2025**
Future Sport & Entertainment (Sydney NSW)

- Evaluated sponsor exposure across sports broadcasts by coding on-screen logo visibility, placement and camera-angle context from image-based datasets.
- Cleaned, validated and standardised Excel datasets for Supercars, Sydney Marathon and Berlin Marathon, supporting downstream automated media-valuation workflows.
- Leveraged TVEyes to track media visibility generated by New Balance-sponsored athletes across international sports coverage.
- Produced client-facing analysis and visualisations on athlete media performance, including US Open coverage of Coco Gauff and Tommy Paul.
- Standardised multinational J.League broadcast datasets, reconciling inconsistent regional formats, time conventions and reporting structures into a unified dataset for analysis and presentation.

Data Analytics Assistant Intern **August 2024 – November 2024**
Sportsyear (Sydney NSW)

- Monitored product usage, booking activity and user engagement across more than 15 partner venues with Google Analytics and Excel.
- Reviewed website, social media and email engagement to identify integration issues and support consistent customer interactions across venue channels.
- Produced regular performance reports and recommendations for internal stakeholders, highlighting trends in product usage and booking behaviour.
- Examined online booking and engagement data for venues including Cheers Bar and The Star Sydney, creating visualisations and insights for stakeholder presentations.

RESEARCH EXPERIENCE

Cyber Risk Modelling & Analytics Researcher
University of Technology Sydney | 2026

Phase 2 — Frequency–Severity Cyber Risk Modelling | MDPI Risks — Accepted

- Developed a system-wide cyber-event frequency forecasting framework from synthetic Hawkes-process event data, comparing statistical and deep-learning approaches including Poisson, Negative Binomial, GRU, LSTM and Transformer-based models.
- Designed rigorous chronological validation and benchmarking procedures to assess model reliability, data leakage, censoring and forecasting limitations.
- Combined probabilistic frequency forecasts with independently calibrated cyber-loss severity distributions to estimate expected aggregate loss, VaR, TVaR and exceedance probabilities.
- Conducted real-data external evaluation with URLhaus to assess how well the models transfer to unseen real-world data.
- Research paper accepted for publication in the MDPI journal Risks.

Phase 1 — Hawkes & Transformer Cyber Risk Prediction | AusDM'26 — Accepted

- Developed and evaluated Hawkes-process, Neural Hawkes, GRU/LSTM and Transformer-based models for multivariate cyber-risk prediction.
- Built Python workflows for event-data preprocessing, model training, validation, benchmarking and performance evaluation.
- Achieved 89.1% accuracy with the Transformer Hawkes Process on the primary top-risk entity prediction task.
- Research project achieved 95/100, High Distinction.
- Research paper accepted for the 2026 Australasian Data Science and Machine Learning Conference (AusDM’26).

EDUCATION

Master of Data Science and Innovation **February 2025 – July 2026**
University of Technology Sydney

- UTS Academic Excellence International Scholarship
- 2026 TD School Merit Award for Outstanding Academic Achievement
- Grade Point Average (GPA): 6.56/7 | Weighted Average Mark (WAM): 87.33/100

Bachelor of Business (Business Analytics) – Distinction **February 2022 – December 2024**
University of Wollongong

- Faculty of Business and Law Country Bursary
- Weighted Average Mark (WAM): 75.5/100

REFEREES

Available Upon Request